

Butian Zhang

Curriculum Vitae

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Research Interests

I work in low-dimensional topology. I am interested in knots, the embedding space of (long) knots and computational knot theory. I study knots from a combinatorial perspective, using tools such as Gauss diagrams, Vassiliev invariants, knot polynomials, and Khovanov homology. A central theme of my research is to understand the topology, especially the homotopy information, of embedding spaces using combinatorial tools. I construct explicit formulae for 0-cocycles and 1-cocycles in the space of (long) knots and study their properties.

Education

2022–2025 **Ph.D. in Mathematics**, *Université de Toulouse*, France.

Advisors: Thomas Fiedler and Stepan Orevkov.

Thesis: **Combinatorial 1-Cocycles of String Links Using Gauss Diagrams**, available at <https://utHEME.utoulouse.fr/s/fr/item/45990>

2020–2022 **M.Sc. in Mathematics (M1 & M2, MINT Program)**, *Université de Toulouse*, France.

2017–2021 **B.Sc. in Mathematics**, *Wuhan University*, China.

Academic Positions

2025–2026 **ATER (Teaching and Research Fellow)**, *Université de Toulouse*, France.

Publications

1. **Gauss diagram formulae for Vassiliev invariants from Kauffman polynomial**, *Journal of Knot Theory and Its Ramifications*, Vol. 32, No. 12, 2350084 (2023). arXiv:2306.01591.

Preprints

1. **Pairings of combinatorial 1-cocycles with loops in knot spaces**, arXiv:2609.29946.
2. **Quantum equations for knots**, with Thomas Fiedler, arXiv:2509.08423.
3. **A combinatorial one-cocycle in a moduli space of knots from the Vassiliev invariant of order 3**, with Baptiste Gros, arXiv:2212.03778.

Talks and Presentations

Aug. 2026 **Moduli and Friends II**, Bucharest.

1-Cohomology Classes from the Long Knot Space to the Closed Knot Spaces

Jun. 2026 **Knotted 3-balls in the 4-sphere**, Matemale.

Spaces of embeddings and the discovery of barbell diffeomorphisms

Apr. 2026 **Oberwolfach Seminar: Knots and Their Interdisciplinary Applications**, MFO, Oberwolfach.

Combinatorial 1-Cocycles in the Space of Long Knots

Apr. 2026 **Virtual Seminar on Discrete Mathematics**, Xiangtan University, Online.

Quantum Equations for Knots

Feb. 2026 **Moscow-Beijing topology seminar**, Online.

Combinatorial 1-Cocycles in the Space of Long Knots

Jan. 2026 **Quantum Topology Seminar**, University of Illinois Chicago, Online.

Combinatorial 1-Cocycles of Order 4 in the Space of Long Knots

- Nov. 2025 **Séminaire des Doctorants, IMT, Toulouse.**
How Do We Distinguish Knots?
- Sep. 2025 **LiT III: Living in Topology, Pisa.**
Combinatorial 1-Cocycles in the Space of Long Knots
- Mar. 2025 **Quid Seminar, IMT, Toulouse.**
Quid of the Space of (Long) Knots
- Feb. 2025 **Winterbraids, IMB, Bordeaux.**
New Combinatorial One-Cocycles of the Space of Long Knots
- May 2024 **Workshop: Exotic 4-Manifolds, Matemale.**
Kirby Calculus and 3-Dimensional Surgery Presentations
- Jul. 2023 **Graduate Student Geometry and Topology Conference, Harvard University, Online.**
Combinatorial One-Cocycles in a Moduli Space of Knots
- May 2023 **Quantum Topology Seminar, University of Illinois Chicago, Online.**
Gauss Diagram Formulae for Vassiliev Invariants from the Kauffman Polynomial

Teaching Experience

- 2025–2026 **Université de Toulouse.**
TP Introduction to Real Analysis
TP Linear Algebra 1
TD Functions and Calculus 1
TD Functions and Calculus 2
- 2024–2025 **Université de Toulouse.**
TP Introduction to Real Analysis
TP Integration and Numerical Series
TD Functions and Calculus 1
- 2023–2024 **Université de Toulouse.**
TP Introduction to Real Analysis
TD Linear Algebra 3
TD Advanced Linear Algebra 3

Skills

- Languages Chinese (native), English, French.
- Programming Python; C/C++

Code

Regular-1-cocycle-HOMFLYPT-2cabled: Python code for computing HOMFLYPT-based regular 1-cocycles. Available at <https://github.com/ZBT99/Regular-1-cocycle-HOMFLYPT-2cabled>.

Finite-type-combinatorial-1-cocycles: Python/C code for computing finite type combinatorial 1-cocycles. Available at <https://github.com/ZBT99/Finite-type-combinatorial-1-cocycles>.